

ARJUNA 2.0
-JEE 2024 Exam-

- Physics
- Kinetic Theory of Gases and Thermodynamics



Lecture No.- 01

By- Himanshu Gupta



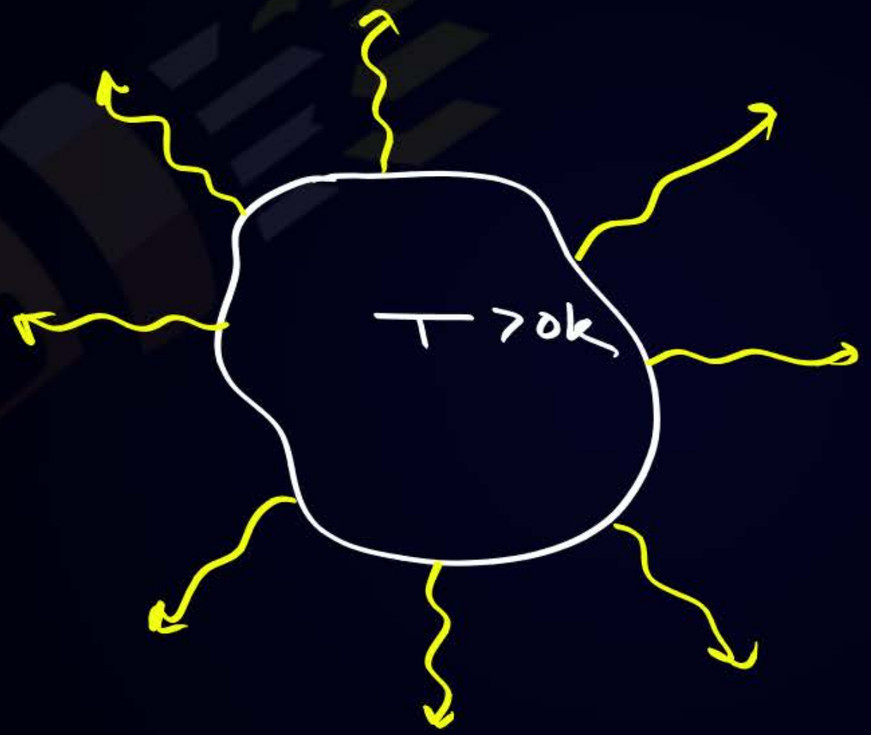
Today's Targets

1. Thermal Radiation
2. Ideal Gas Assumptions

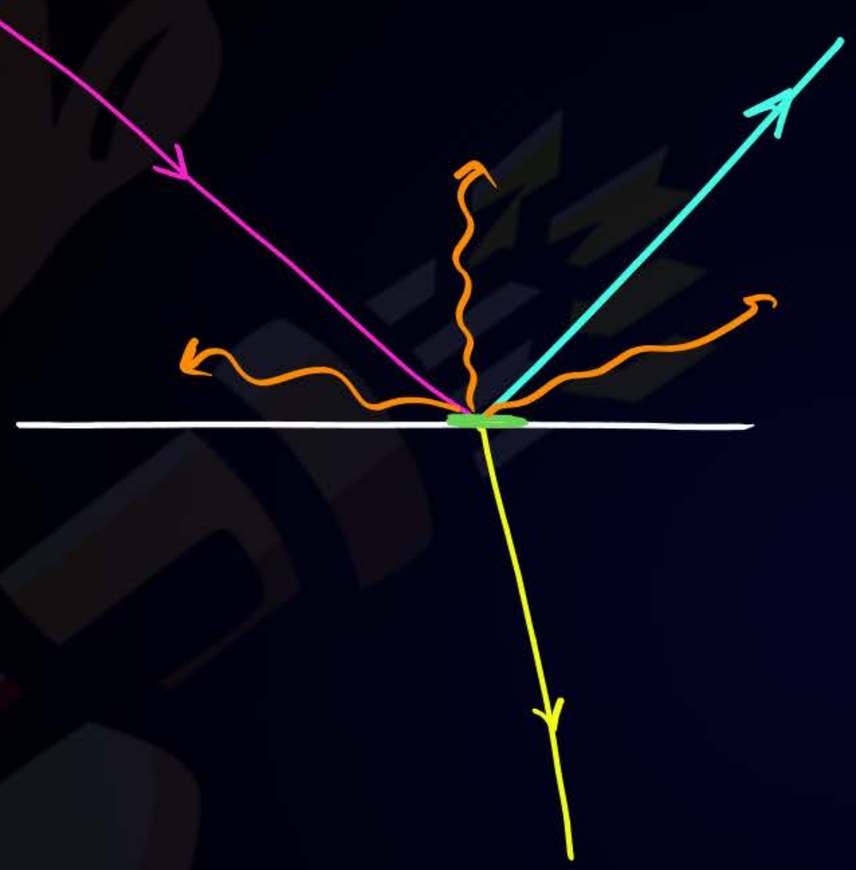
Test date :- 5-th Jan.

Quick Revision

Thermal Radiation



Prevost Theory



Reflectivity, $r = \dfrac{\text{Reflected Energy}}{\text{Incident Energy}}$

Transmittivity $t = \dfrac{\text{Transmitted Energy}}{\text{Incident Energy}}$

Absorptivity, $a = \dfrac{\text{Absorbed Energy}}{\text{Incident Energy}}$

Absorptive Power

$$r + t + a = 1$$

Emissive Power, Absorptive Power

Absorptive Power, absorptivity, $a = \frac{\text{absorbed Energy}}{\text{Incident Energy}}$

Emissive Power, $E = \frac{\text{Energy Radiated}}{\text{time} \times \text{Area}}$

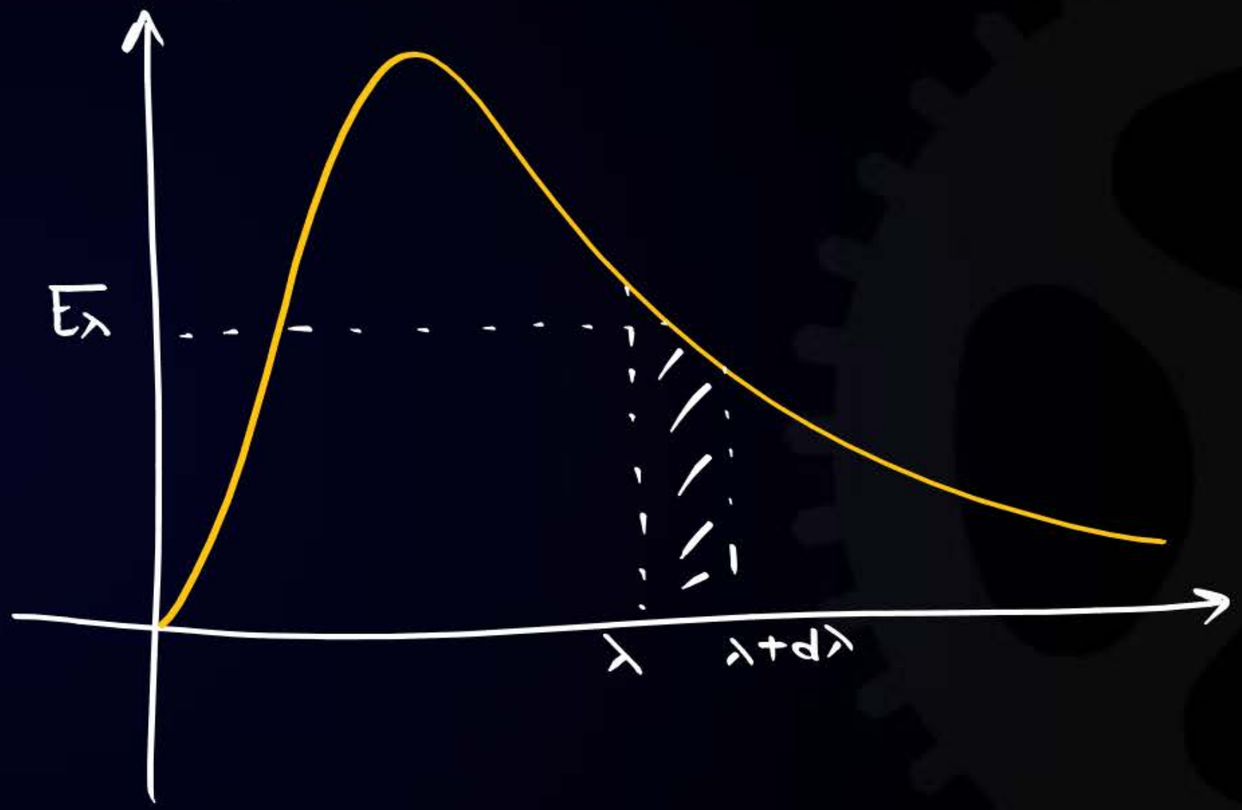
Spectral Emissive Power

$$E_\lambda = \frac{E}{\lambda}$$

$$E_\lambda = \frac{dE}{d\lambda}$$

$$E_\lambda = \frac{\text{Energy}}{\text{Time} \times \text{Area} \times \lambda}$$

$$\text{Total rad. Energy} = \int E_\lambda \, d\lambda$$

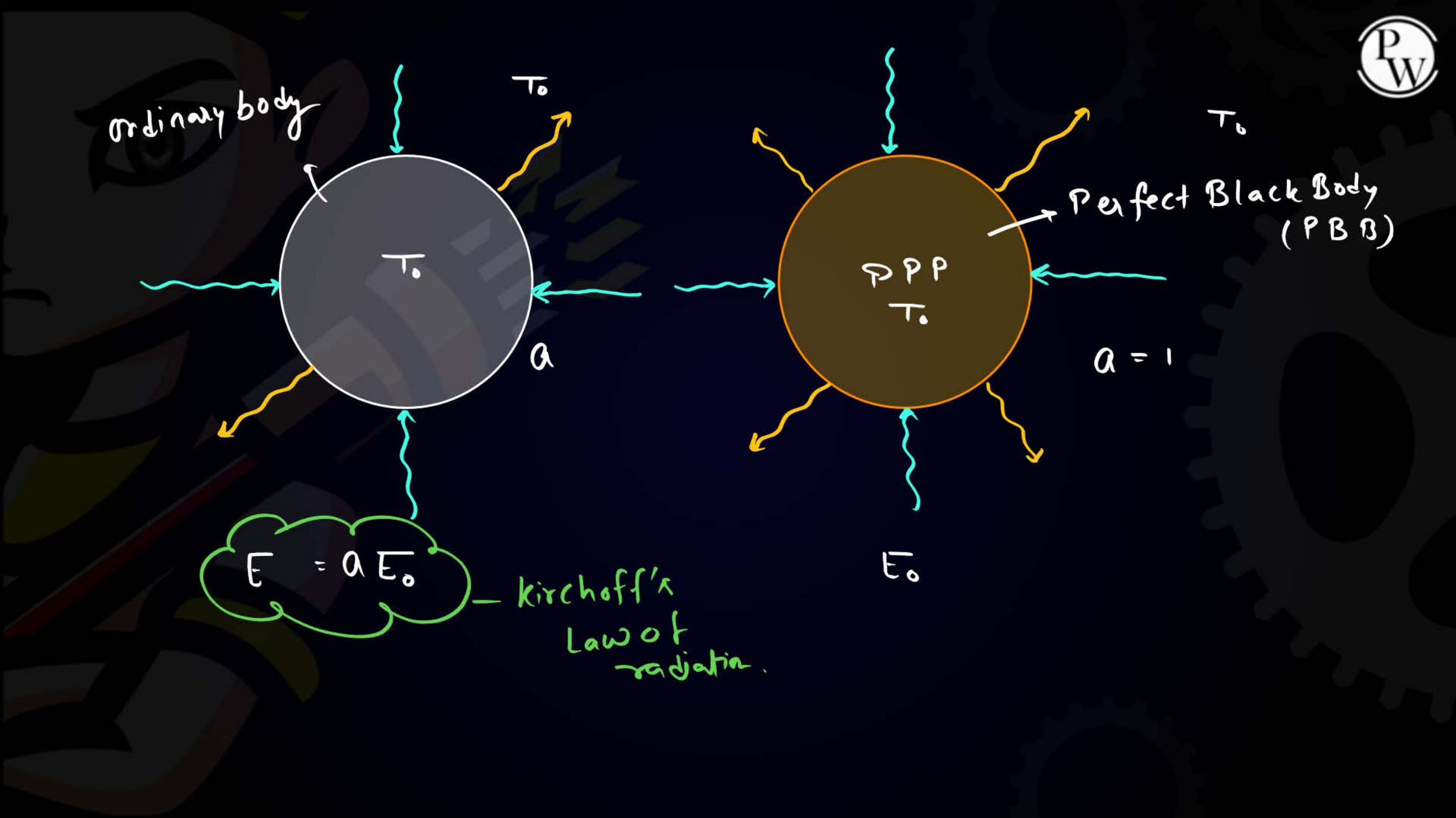


Perfect Black Body

$a = 1, \quad r = 0, \quad t = 0$

a ↑ absorptivity




ordinary body
T_0
T_0
a
$E = aE_0$
Kirchoff's Law of radiation.
Perfect Black Body (PBB)
T_0
$a = 1$
E_0

Kirchhoff's Law of Radiation

$E = a E_o$ → $\frac{E}{E_o} = a$ → Absorptive power

Emissive power of an ordinary body is a fraction of Emissive power of PBB at same temp.

E = Emissive power

E_o = Emissive power of PBB at same temp.

Stefan-Boltzmann Law

(Experimentally measured)

$$\frac{dQ}{dt} = \sigma e A T^4$$



$e = 1$ for PBB.

$\frac{dQ}{dt}$ = Heat rad. by body per sec

σ = Stefan's Const. = $5.67 \times 10^{-8} \frac{W}{m^2 K^4}$

e = Emissivity $(0 < e \leq 1)$

A = Surface area of body

T = Absolute temp. of body (in kelvin)

QUESTION (JEE Mains – 2006)

Assuming the Sun to be a spherical body of radius R at a temperature of TK, evaluate the total radiant powered incident ~~of~~ on Earth at a distance r from the Sun

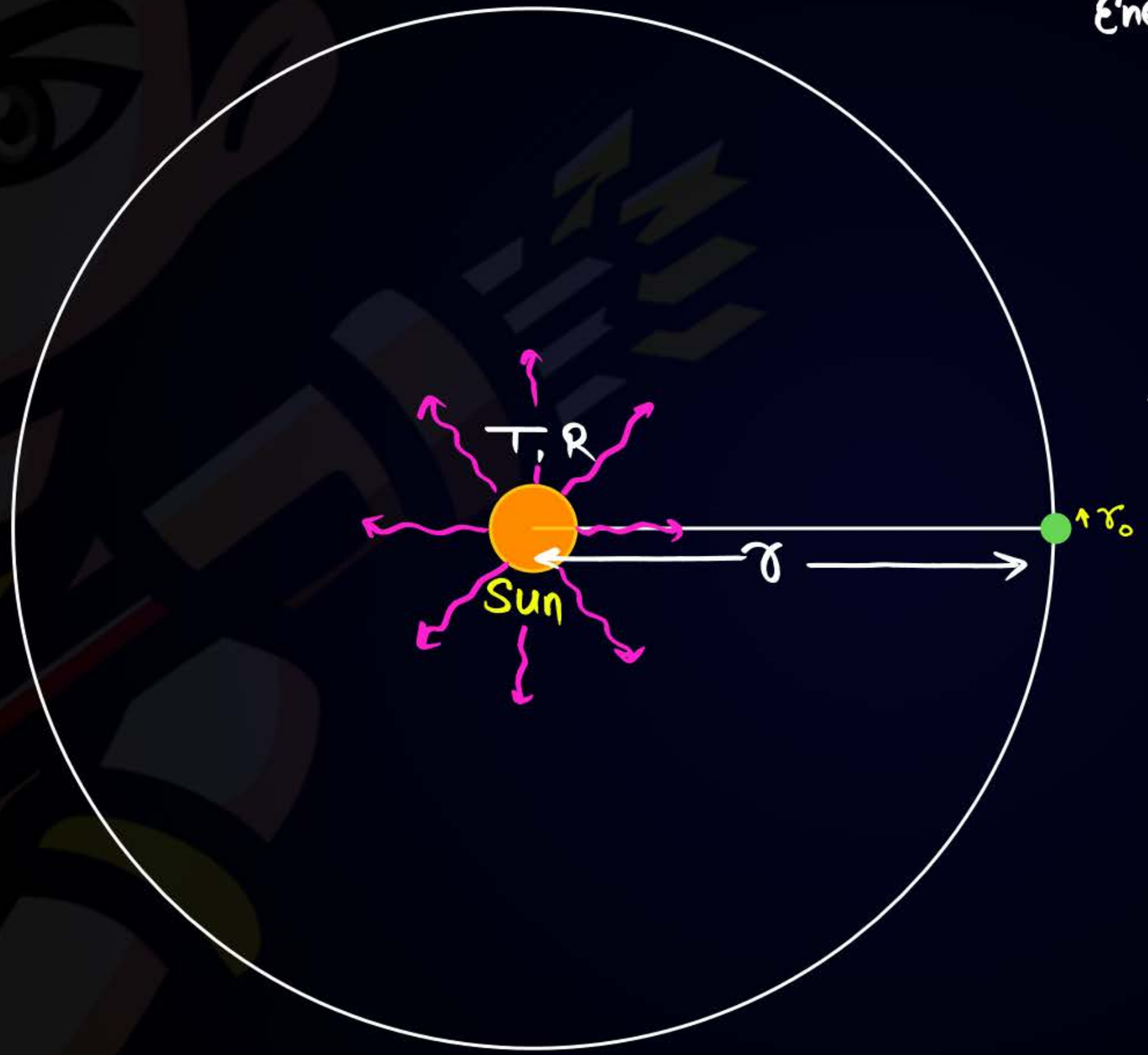
A $4\pi r_0^2 R^2 \sigma \frac{T^4}{r^2}$

B $\pi r_0^2 R^2 \sigma \frac{T^4}{r^2}$

C $r_0^2 R^2 \sigma \frac{T^4}{4\pi r^2}$

D $R^2 \sigma \frac{T^4}{r^2}$

where r_0 is the radius of the Earth and σ is Stefan's constant.



Energy rad. per sec. by sun —

$$\frac{dQ}{dt} = \sigma e A T^4$$

$$\frac{dQ}{dt} = \sigma \cdot 1 \cdot (4\pi R^2) T^4$$

$$\frac{dQ}{dt} = 4\pi\sigma R^2 T^4$$

Energy received by Earth —

$$= \frac{\frac{dQ}{dt}}{4\pi r^2} \times \pi r_0^2$$

$$= \frac{4\pi\sigma R^2 T^4 r_0^2}{4r^2}$$

QUESTION (JEE Mains – Online May 7, 2012)

The heat radiated per unit area in 1 hour by a furnace whose temperature is 3000 K is ($\sigma = 5.7 \times 10^{-8}$ W m^{-2} K^{-4})

A. 1.7×10^{10} J

B. 1.1×10^{12} J

C. 2.8×10^{8} J

D. 4.6×10^{6} J

$$\frac{dQ}{dt} = \sigma e A T^4$$

$$\frac{dQ}{dt} = 5.67 \times 10^{-8} \times 1 \times 1 \times (3000)^4$$

$$Q = 5.67 \times 10^{-8} \times 81 \times 10^{12} \times 3600$$

$$= 5.67 \times 81 \times 36 \times 10^6$$

$$= 1.7 \times 10^{10}$$

QUESTION (JEE Mains – 2004)

If the temperature of the sun were to increase from T to $2T$ and its radius from R to $2R$, then the ratio of the radiant energy received on earth to what it was previously will be

A 32

B 16

C 4

D 64 ✓

$$\frac{dQ}{dt} = \sigma e A T^4$$

$$\frac{dQ}{dt} = \sigma e (4A) \times (2T)^4$$

$$= 4 \times 16 \, \sigma e A T^4$$

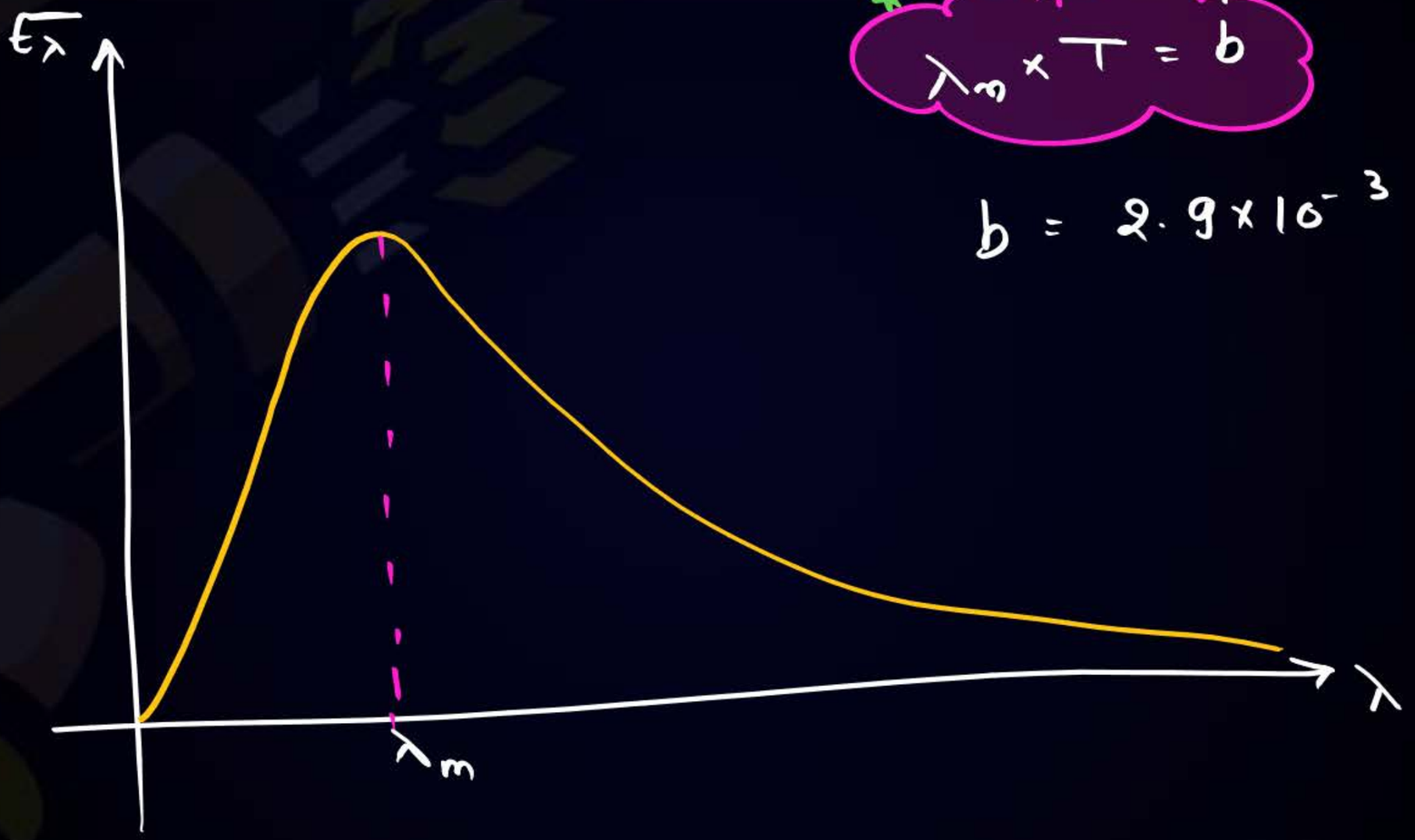
$$= 64$$

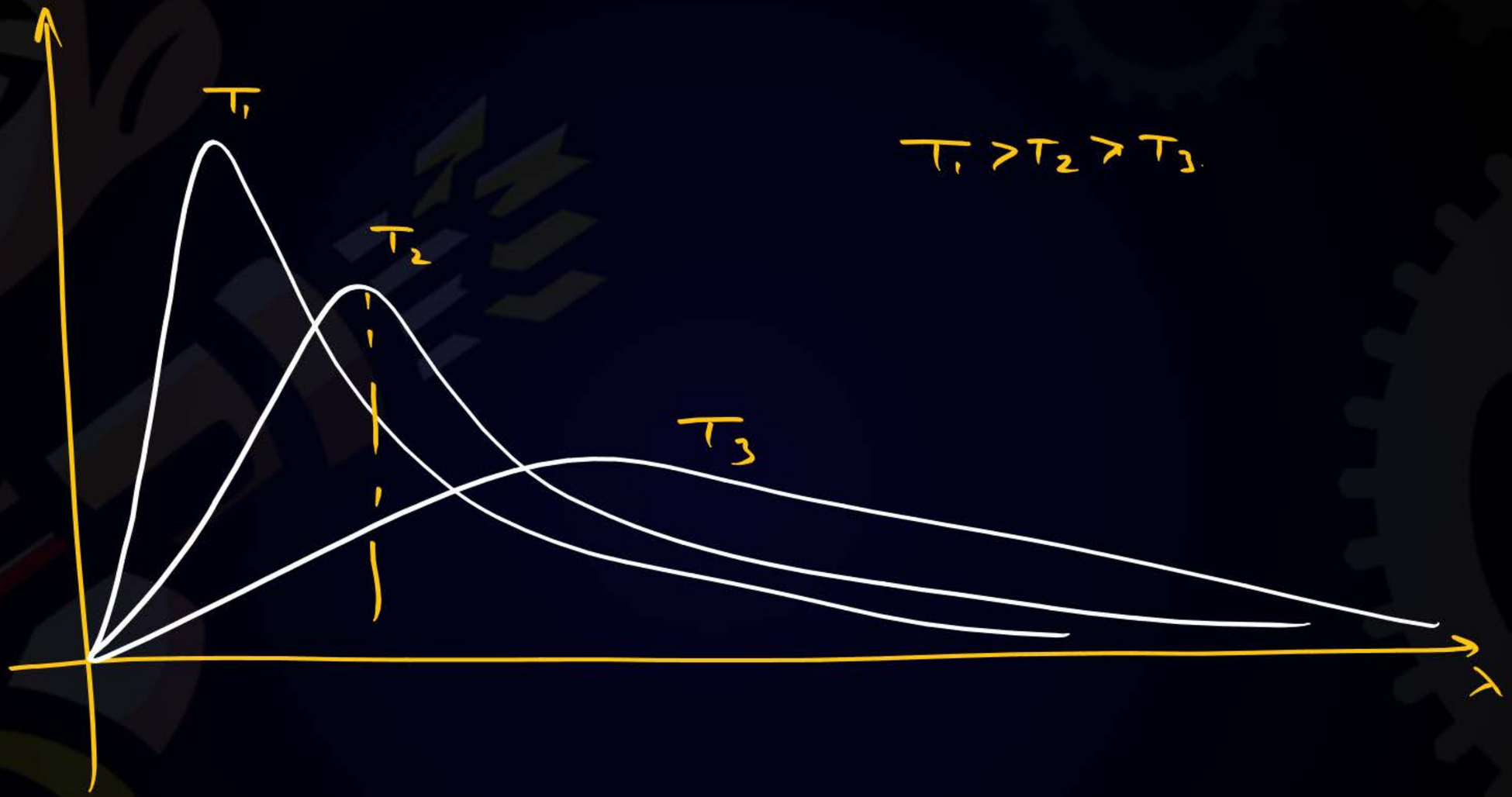
Wein's Displacement Law

Const

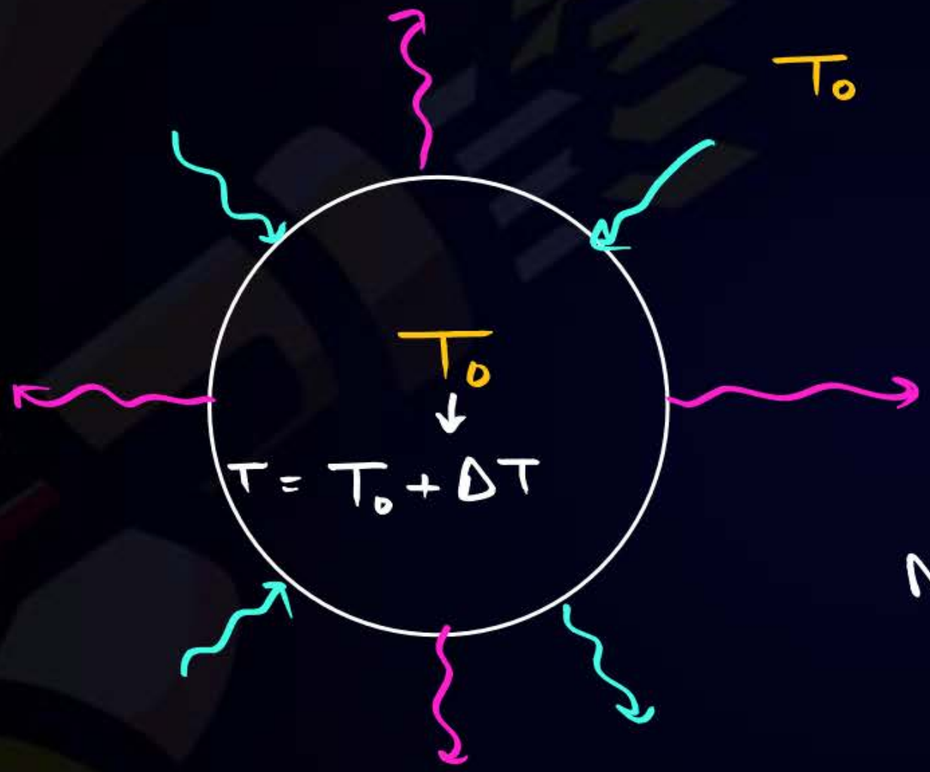
$\lambda_m \times T = b$

$b = 2.9 \times 10^{-3}$ mk.




T_1
T_2
T_3
$T_1 > T_2 > T_3$
λ

Rate of Heat Loss



At temp. $T = T_0 \rightarrow$

Heat absorbed per sec. = Heat radiated by body per sec

Heat absorbed per sec $= \sigma e A T_0^4$

When body's temp. is raised by ΔT –

Net rate of Heat loss = Heat rad/sec − Heat absorbed per sec

$= \sigma e A T^4 - \sigma e A T_0^4$

$= \sigma e A (T^4 - T_0^4)$

$$\frac{dQ}{dt} = \sigma e A \left[T^4 - T_0^4\right]$$

$$\frac{dQ}{dt} = \sigma e A \left[(T_0 + \Delta T)^4 - T_0^4\right]$$

$$\frac{dQ}{dt} = \sigma e A \left[T_0^4 \left(1 + \frac{\Delta T}{T_0}\right)^4 - T_0^4\right]$$

$$\frac{dQ}{dt} = \sigma e A \left[T_0^4 \left(1 + 4\frac{\Delta T}{T_0}\right) - T_0^4\right]$$

$$\frac{dQ}{dt} = \sigma e A \left[4 T_0^3 \Delta T\right]$$

$$\frac{dQ}{dt} = 4\sigma e A T_0^3 \Delta T$$

$$\downarrow$$

$$-ms\frac{dT}{dt} = 4\sigma e A T_0^3 (T - T_0)$$

$$\frac{-dT}{dt} = \underbrace{\frac{4\sigma e A T_0^3}{ms}}_{k}(T - T_0)$$

$$\frac{-dT}{dt} = k(T - T_0)$$

Rate of cooling

Newton's Law of Cooling

$$-\frac{dT}{dt} = k(T - T_0)$$

$-\frac{dT}{dt}$ = Rate of cooling

T = Temp. of body

T_0 = temp. of surrounding

$$-\frac{dT}{dt} = k(T - T_0)$$

$$\int_{T_i}^{T_f} \frac{dT}{T - T_0} = \int_0^t -k\, dt$$

$$\left| \ln(T - T_0) \right|_{T_i}^{T_f} = -kt$$

$$\ln \frac{T_f - T_0}{T_i - T_0} = -kt$$

$$\Rightarrow \ln\left(\frac{T_i - T_0}{T_f - T_0}\right) = kt$$

Question

If $80°C \rightarrow 70°C = 100s$ then $70°C \rightarrow 60°C$, $t = ?$

$80^\circ C \longrightarrow 70^\circ C$: 100 sec.

$70^\circ C \longrightarrow 60^\circ C$: $t = ?$

$T_0 = 20^\circ C$

$$\frac{\ln\left(\frac{80-20}{70-20}\right) = k \times 100}{\ln\left(\frac{70-20}{60-20}\right) = k \times t}$$

$$\rightarrow \frac{\ln \frac{6}{5}}{\ln \frac{5}{4}} = \frac{100}{t}$$

$$t = \frac{100 \ln \frac{5}{4}}{\ln 6/5}$$

Brahmastra

$T_0 = 20^\circ C$

$80^\circ C \longrightarrow 70^\circ C : 100 sec.$

$70^\circ C \longrightarrow 60^\circ C : t$

$$\frac{\Delta T}{\Delta t} = -k(T_{avg} - T_0)$$

$$\frac{10^\circ C}{100 sec} = +k\left[\left(\frac{80+70}{2}\right) - 20\right]$$

$$\frac{10^\circ C}{t} = +k\left[\left(\frac{70+60}{2}\right) - 20\right]$$

$$\frac{t}{100} = \frac{75-20}{65-20}$$

$$t = 100 \times \frac{55}{45}$$ Ans.

QUESTION (JEE Mains – Sep. 03, 2020 (II))

HW-1

A metallic sphere cools from 50°C to 40°C in 300 s. If atmospheric temperature around is 20°C, then the sphere's temperature after the next 5 minutes will be close to :

A 31°C

B 33°C

C 28°C

D 35°C

QUESTION (JEE Mains – July. 27, 2021 (I))

H·W·2

A body takes 4 min. to cool from 61°C to 59°C. If the temperature of the surroundings is 30°C, the time taken by the body to cool from 51°C to 49°C is:

A 6 min

B 3 min

C 4 min

D 8 min

QUESTION (JEE Mains – 2014)

H·W·3

Hot water cools from 60°C to 50°C in the first 10 minutes and to 42°C in the next 10 minutes. The temperature of the surroundings is:

A 25°C

B 10°C

C 15°C

D 20°C

QUESTION (JEE Mains – JEE Adv, 2020)

A container with 1 kg of water in it is kept in sunlight, which causes the water to get warmer than the surroundings. The average energy per unit time per unit area received due to the sunlight is 700 Wm^{-2} and it is absorbed by the water over an effective area of 0.05 m^2. Assuming that the heat loss from the water to the surroundings is governed by Newton's law of cooling, the difference (in °C) in the temperature of water and the surroundings after a long time will be _____. (Ignore effect of the container, and take constant for Newton's law of cooling = 0.001s^{-1}, Heat capacity of water = 4200 J kg^{-1} K^{-1})

KTG & Thermodynamics



Ideal Gas Assumptions

1. All gases are made of molecules moving randomly in all directions.
2. The size of a molecule is much smaller than the average separation between the molecules.
3. The molecules exert no force on each other or on the walls of the container except during collision.
4. All collisions between two molecules or between a molecule and a wall are perfectly elastic. Also, the time spent during a collision is negligible small.

5. The molecules obey Newton's Laws of Motion.
6. When a gas is left for sufficient time, it comes to a steady state. The density and the distribution of molecules with different velocities are independent of position, direction and time. This assumption may be justified if the number of molecules is very large.


COUPONS

QUESTION (JEE Mains – Apr. 13, 2023 (II))

H.W-1

Two plates A and B have thermal conductivities 84 $Wm^{-1}K^{-1}$ and 126 $Wm^{-1}K^{-1}$ respectively. They have same surface area and same thickness. They are placed in contact along their surfaces. If the temperatures of the outer surfaces of A and B are kept at 100° C and 0°C respectively, then the temperature of the surface of contact in steady state is _____ °C.



$$R_1 + R_2 = \frac{l}{A}\left[\frac{1}{84} + \frac{1}{126}\right]$$

$$= \frac{l}{A}\left[\frac{3+2}{252}\right]$$

$$R_1 + R_2 = \frac{5l}{252A}$$

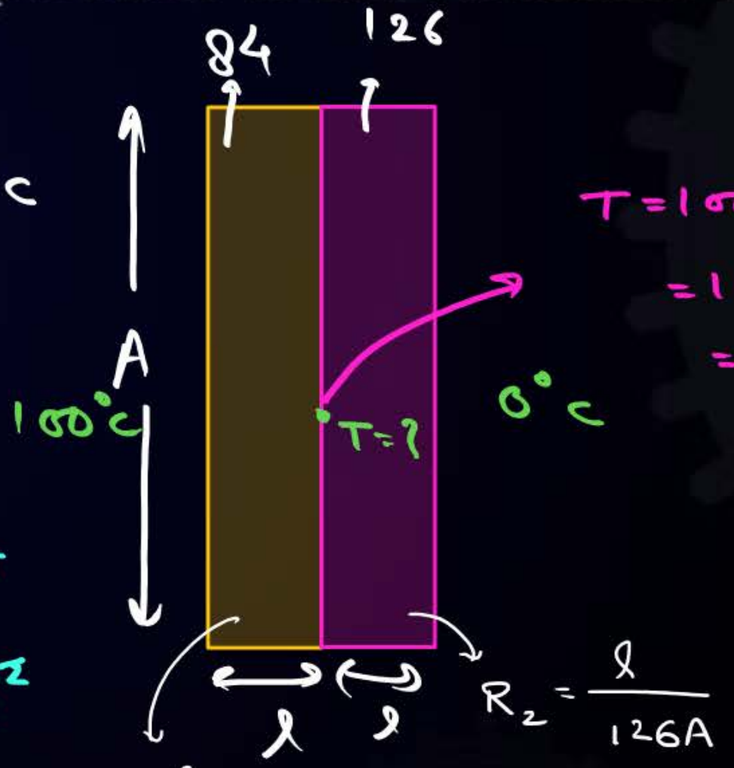
$R_1 + R_2 \longrightarrow 100°C$

$1 \longrightarrow \frac{100}{R_1 + R_2}$

$R_1 \longrightarrow \frac{100}{R_1 + R_2} * R_1 = \Delta T$

$\frac{100}{\left(\frac{5l}{252A}\right)} * \frac{l}{84A} = \frac{100 \times 252}{5 \times 84}$

$= 60°C$



$T = 100 - \Delta T$

$= 100 - 60$

$= 40°C$ Ans.

Homework

DPP:	NO DPP (Th. DPP)
MODULE:	

Revision Test No.23:	COM
(27 Dec 2023, Wednesday)	Lectures – 1, 2, 3, 4, 5, 6, 7


THANK YOU